

Problem Set 7: Poisons, Rods, and Flux Shaping

Due: Friday, March 20, 2026

Instructions

- Show all work, including the equations used and unit conversions.
 - Assume standard LWR constants unless otherwise stated:
 - Effective Delayed Neutron Fraction: $\beta = 0.0065$
 - Iodine-135 Decay Constant: $\lambda_I = 2.9 \times 10^{-5} \text{ s}^{-1}$ ($t_{1/2} = 6.6 \text{ hr}$)
 - Xenon-135 Decay Constant: $\lambda_X = 2.1 \times 10^{-5} \text{ s}^{-1}$ ($t_{1/2} = 9.1 \text{ hr}$)
 - Xenon-135 Absorption Cross-section: $\sigma_a^X = 2.6 \times 10^6 \text{ barns}$
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1. The Reactivity Cost of Power (Equilibrium Xenon)

A reactor operates at a steady thermal flux of $\phi = 10^{13} \text{ n/cm}^2\text{s}$. The equilibrium concentration of Xenon-135 is given by:

$$X_\infty = \frac{(\gamma_I + \gamma_X)\Sigma_f\phi}{\lambda_X + \sigma_a^X\phi}$$

- (a) Calculate the term $\sigma_a^X\phi$ (the burnout rate) and compare it to λ_X (the decay rate). Which removal mechanism dominates at this flux level?
- (b) If the reactor power is increased such that the flux doubles to $\phi = 2 \times 10^{13} \text{ n/cm}^2\text{s}$, by what factor does the equilibrium Xenon concentration change? (Does it double, stay the same, or something else?)
- (c) Calculate the macroscopic absorption cross-section of the Xenon ($\Sigma_a^X = \sigma_a^X X_\infty$) and the resulting negative reactivity worth $\rho_{Xe} \approx -\frac{\Sigma_a^X}{\Sigma_a^{Total}}$. (Assume $\Sigma_f/\Sigma_a^{Total} \approx 1/1.2$ and total yield $\gamma \approx 0.06$).

2. The "Dead Time" After Shutdown

A reactor scrams after a long run at full power (100%).

- (a) Explain physically why the Xenon-135 concentration *rises* immediately after shutdown, even though no new fission is occurring.
- (b) The "Peak Xenon" occurs approximately 11 hours after shutdown. If the reactor has a "Shutdown Margin" of only 5% $\Delta k/k$, and the Xenon peak adds $-8\% \Delta k/k$ of negative reactivity, what is the operational implication for the plant staff?
- (c) How does the magnitude of this "Iodine Pit" differ for a reactor that scrammed from 50% power versus one that scrammed from 100% power?

3. The S-Curve Calculation

Consider a control rod with a total worth of $\rho_{total} = \$2.00$ (2 dollars). The reactor height is $H = 400$ cm. The integral worth of the rod as a function of withdrawal position z is given by:

$$\rho(z) = \rho_{total} \left[\frac{z}{H} - \frac{1}{2\pi} \sin \left(\frac{2\pi z}{H} \right) \right]$$

- Calculate the reactivity inserted when the rod is withdrawn from 0 cm to 100 cm (the first 25% of travel).
- Calculate the reactivity inserted when the rod is withdrawn from 150 cm to 250 cm (the middle 25% of travel).
- Compare the two values. Why is the second value so much larger? (Reference the flux shape in your answer).

4. The Hot Spot Factor (F_q)

In a bare cylindrical reactor, the power distribution follows the shape:

$$P(r, z) = P_{max} J_0 \left(\frac{2.405r}{R} \right) \cos \left(\frac{\pi z}{H} \right)$$

The Peaking Factor F_q is the ratio of the maximum power density to the average power density.

- For this "natural" shape, the theoretical F_q is approximately 3.64 (radial component $2.32 \times$ axial component 1.57). If the reactor produces 3000 MW_{th} total, and the core volume is 30 m³, what is the **Average** Power Density (MW/m³)?
- What is the **Peak** Power Density (MW/m³) at the center of the core?
- An engineer proposes using "Zone Loading" (putting lower enrichment fuel in the center) to flatten the flux, reducing F_q to 2.2. If the safety limit for the fuel is 350 MW/m³, could we increase the total power of the reactor with this new design? If so, by how much?

5. The Stable Poison Problem

Unlike Xenon-135, Samarium-149 is a stable isotope.

- A reactor operates for 6 months and then shuts down for maintenance. Xenon-135 decays away within 3 days. What happens to the Samarium-149 concentration during this shutdown?
- When the reactor restarts, how does the behavior of Samarium differ from Xenon? (Hint: Does Samarium have a "peak"? Does it burnout?)
- Why is the "Promethium Dip" (the Samarium equivalent of the Iodine Pit) usually not an operational safety concern during startup, whereas the Xenon peak is?